MicrobiomeExperiment and the emerging R ecosystem for microbiome research

PRESENTER INFO:

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MORE INFO @ EuroBioc2020:

Short Talk: Wednesday, 16:40 - 16:50 CET
Workshop: Friday, 17:05 - 18:05 CET

CONTRIBUTING AUTHORS:

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Check out the miaBook on GitHub:

MOTIVATION

- Existing microbiome workflows in Bioconductor are largely unsupported by the rich and versatile SummarizedExperiment (`SE`) family of classes.
- Recently contributed packages now provide tools for multiomic and time-series analyses which work alongside both new and old tools for metagenomic analyses.

PROPOSAL

- We propose a pipeline using `TreeSummarizedExperiment` (`TreeSE`) and the Microbiome Analysis package (`mia`) for computational metagenomics with multi-omic data.
- We hope to reduce overlapping efforts, improve interoperability and ensure the long-term sustainability of microbiome research for the R and Bioconductor community.

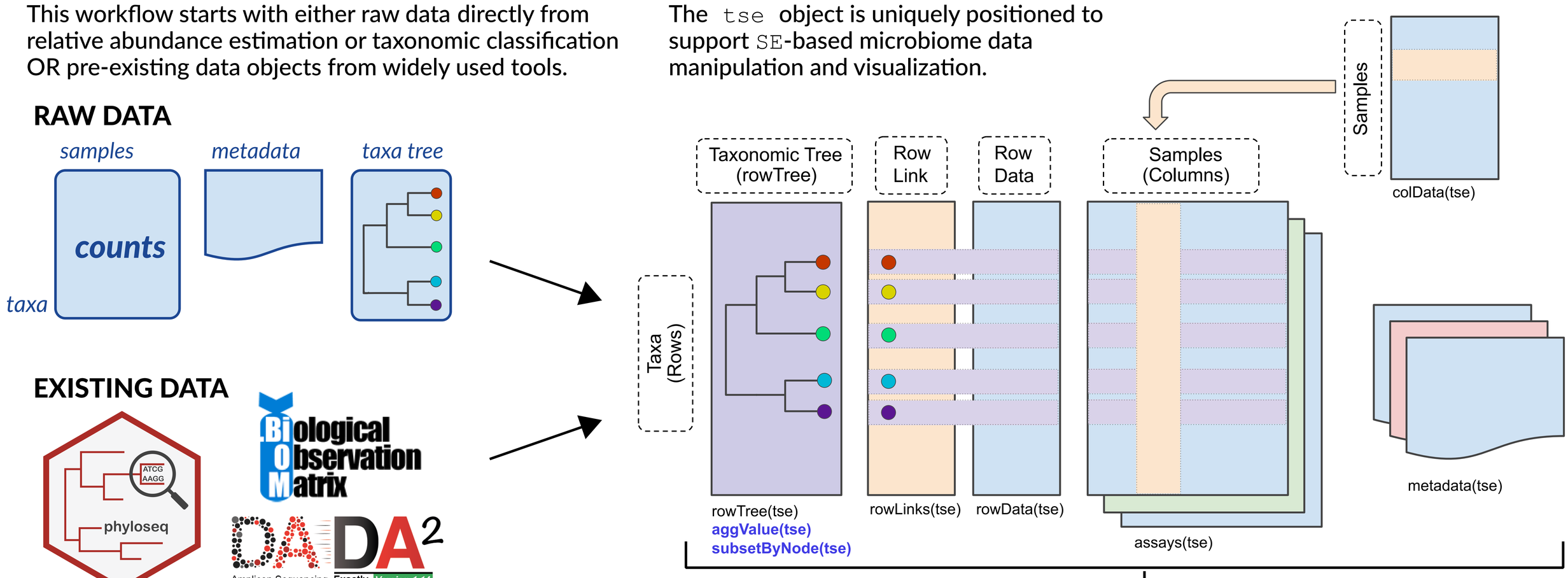
PROPOSED WORKFLOW: `TreeSE` & `mia`

Import Data

This workflow starts with either raw data directly from relative abundance estimation or taxonomic classification OR pre-existing data objects from widely used tools.

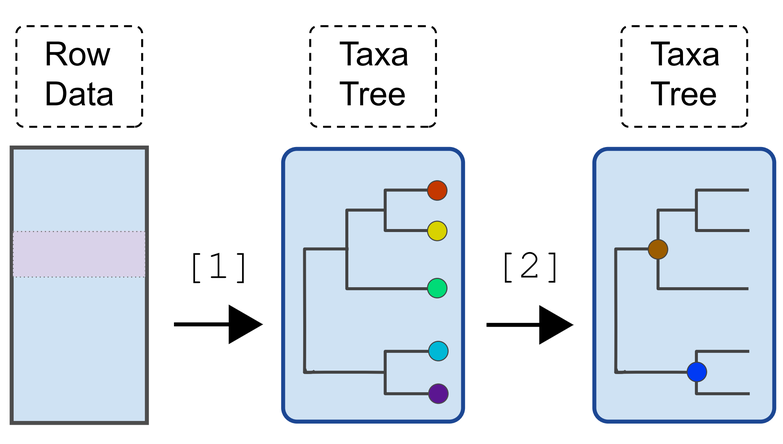
The `TreeSE` object

The `tse` object is uniquely positioned to support `SE`-based microbiome data manipulation and visualization.



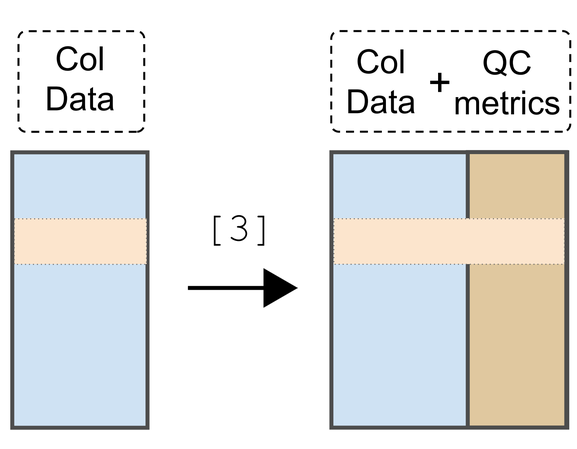
The `mia` Pipeline

Accessing Taxonomic Info.



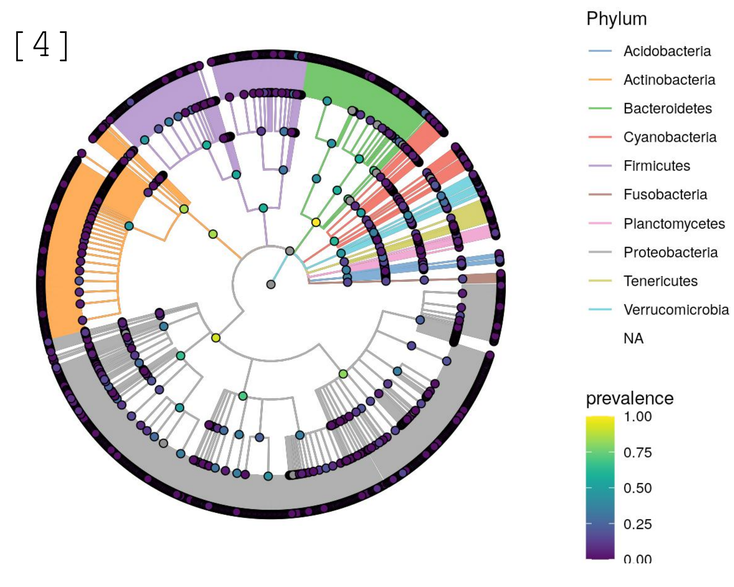

```
[1] mia::addTaxonomyTree(tse)
[2] TreeSE::aggValue(tse)
```

Quality Control




```
[3] scater::addPerCellQC(tse)
```

Visualizing with `miaViz`




```
[4] miaViz::plotRowTree(tse)
```